

Solutions and Solubility — Paper C

HARD • NON-CALCULATOR • 45 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided.
- No calculator is needed anywhere on this paper.
- Where a question says **explain**, your answer must talk about **particles** — solute particles spreading between solvent particles, collisions, or how much the solvent can hold at that temperature.
- Solubility is always given in **g per 100 g of water**. If a question uses a different mass of water, scale it.
- The mark for each question is shown in square brackets on the right.
- Several questions here describe experiments you may not have seen before. You are not expected to recognise them — apply the ideas you already have.
- Show every subtraction. A correct number with no working is one mark, not three.

SECTION 1 – AMOUNT AGAINST SPEED

- 1.** A student stirs a saturated sugar solution for ten minutes, expecting the sugar on the bottom to dissolve. It does not.

(a) Explain why stirring did not dissolve the remaining sugar. [2]

(b) State what stirring does change, and explain it in terms of particles. [2]

- 2.** Two students each stir 50 g of sodium chloride into 100 g of water. Priya uses water at 20 °C; Arjun uses water at 80 °C. Solubility of sodium chloride: 36 g per 100 g at 20 °C, 38 g per 100 g at 80 °C.

(a) Calculate the mass left undissolved in each beaker. [2]

- (b)** Arjun says heating "made hardly any difference". Use the data to explain why he is right about the amount. [1]

- (c)** Arjun's salt dissolved much faster than Priya's. Explain why, in terms of particles. [2]

SECTION 2 – GASES

- 3.** A pond warms up in summer and fish are seen gasping at the surface.

- (a)** State how the solubility of oxygen in water changes as the water warms. [1]

- (b)** Explain why, in terms of particles. [2]

SECTION 3 – CALCULATIONS AND PLANNING

- 4.** A saturated solution of potassium nitrate is made in **50 g** of water at 60 °C and cooled to 20 °C. Solubility: 110 g per 100 g at 60 °C; 32 g per 100 g at 20 °C.

- (a)** Calculate the mass dissolved at 60 °C. [2]

- (b)** Calculate the mass of crystals that forms on cooling to 20 °C. Show your working. [3]

5. A saturated solution of copper sulfate is made in **200 g** of water at 80 °C and cooled to 20 °C. Solubility: 55 g per 100 g at 80 °C; 21 g per 100 g at 20 °C. Calculate the mass of crystals that forms. Show your working. [3]

6. A student wants to find out whether sugar is more soluble in water than salt is, at room temperature. Describe a fair test she could do. Say what she would measure, what she must keep the same, and what result would show sugar is more soluble. [5]

7. Explain why a dilute solution can still be a saturated solution. [2]

8. A beaker of salt solution is left on a balance for a week and the reading falls by 12 g. A student says "some of the salt has disappeared". Explain what has really happened. [3]

End of paper. Check every answer has a unit or a form the question actually asked for.